DONG-HO KANG

Associate Professor, Ph.D.
Department of Semiconductor Engineering
Department of Electrical Engineering and Computer Science (EECS)
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EDUCATION

M.S. / Ph.D. Sungkyunkwan University — 2019
Electronic & Electrical Engineering
Dissertation: Process Technology for 2D Semiconducting Devices *(Advisor: Prof. Jin-Hong Park)*

B.S. Sungkyunkwan University — 2014
Electronic & Electrical Engineering

APPOINTMENTS

Department Chair — Mar. 2026 – Present
Gwangju Institute of Science & Technology
Department of Semiconductor Engineering

Associate Professor — Mar. 2026 – Present
Gwangju Institute of Science & Technology
Department of Semiconductor Engineering

Assistant Professor — Mar. 2025 – Feb. 2026
Gwangju Institute of Science & Technology
Department of Semiconductor Engineering

Assistant Professor — Nov. 2021 – Feb. 2025
Gwangju Institute of Science & Technology
School of Electrical Engineering and Computer Science

Research Fellow — Mar. 2019 – Sep. 2021
Nanyang Technological University
School of Electrical and Electronic Engineering

RESEARCH INTEREST

Logic Devices — Process technologies for 2D materials — from synthesis and transfer to doping and junction control — toward next-generation logic devices.

Optoelectronic Devices — Neuromorphic and optoelectronic synaptic devices that emulate synapses and neurons for next-generation AI semiconductors.

System-level AI Hardware — In-sensor computing and spiking neural network hardware that realize AI semiconductors at the system level.

AWARDS

Young Scientist Award 2025
Korea Flexible & Printed Electronics Society(한국유연인쇄전자학회)

Superior Research Award (Ph.D. Course), Second Prize 2017
Sungkyunkwan University
Electrical and Electronic Sector

Superior Research Award (M.S. Course), First Prize 2015
Sungkyunkwan University
Electrical and Electronic Sector

Superior Research Award (M.S. Course), First Prize 2014
Sungkyunkwan University
Electrical and Electronic Sector

SELECTED PUBLICATIONS

Top 10% Journal (JCR)

No.	Title	Journal	Authors
8	Revisiting Ferroelectric-Gated Phototransistors: A Tripartite Synapse-Inspired Approach to In-Sensor Image Processing	***Adv. Mater.*** (2026)	Y. Lee, D. H. Seo+, J. S. Lee+, J. M. Jeon+, H. R. Kim, M. S. Kim, C. Ahn, S.-U. An, J. Choi, H. Kim, C. K. Jeong, H. Lin, **D.-H. Kang***, Y. M. Song*
7	A Van der Waals Optoelectronic Synapse With Tunable Positive and Negative Post-Synaptic Current for Highly Accurate Spiking Neural Networks	***Adv. Funct. Mater.*** (2026)	H. Yoon+, S. Park+, Y. K. Kim+, J. Baek, K. H. Kim, S. Yun, H. Son, J. Choi, B. C. Jang*, **D.-H. Kang***
6	Tactile neuromorphic system: convergence of triboelectric polymer sensor and ferroelectric polymer synapse	***ACS Nano*** (2023)	H. Kim, S. Oh, H. Choo, **D.-H. Kang***, J.-H. Park*
5	Pseudo-magnetic field-induced slow carrier dynamics in periodically strained graphene	***Nat. Commun.*** (2021)	**D.-H. Kang**+, H. Sun+, M. Luo+, K. Lu, M. Chen, Y. Kim, Y. Jung, X. Gao, S. J. Parluhutan, J. Ge, S. W. Koh, D. Giovanni, T. C. Sum, Q. J. Wang, H. Li, D. Nam*
4	A High-Performance WSe2/h-BN Photodetector using a Triphenylphosphine (PPh3)-Based n-Doping Technique	***Adv. Mater.*** (2016)	S.-H. Jo+, **D.-H. Kang**+, J. Shim, J. Jeon, M. H. Jeon, G. Yoo, J. Kim, J. Lee, G. Y. Yeom, S. Lee, H.-Y. Yu, C. Choi, J.-H. Park*
3	An ultrahigh-performance photodetector based on a perovskite--transition-metal-dichalcogenide hybrid structure	***Adv. Mater.*** (2016)	**D.-H. Kang**, S. R. Pae, J. Shim, G. Yoo, J. Jeon, J. W. Leem, J. S. Yu, S. Lee, B. Shin*, J.-H. Park*
2	Controllable nondegenerate p-type doping of tungsten diselenide by octadecyltrichlorosilane	***ACS Nano*** (2015)	**D.-H. Kang**, J. Shim, S. K. Jang, J. Jeon, M. H. Jeon, G. Y. Yeom, W.-S. Jung, Y. H. Jang, S. Lee, J.-H. Park*
1	High-performance transition metal dichalcogenide photodetectors enhanced by self-assembled monolayer doping	***Adv. Funct. Mater.*** (2015)	**D.-H. Kang**+, M.-S. Kim+, J. Shim, J. Jeon, H.-Y. Park, W.-S. Jung, H.-Y. Yu, C.-H. Pang, S. Lee, J.-H. Park*

FULL-PUBLICATION LIST (MAIN-AUTHORED)

Top 10% Journal (JCR)

No.	Title	Journal	Authors
23	Revisiting Ferroelectric-Gated Phototransistors: A Tripartite Synapse-Inspired Approach to In-Sensor Image Processing	***Adv. Mater.*** (2026)	Y. Lee, D. H. Seo+, J. S. Lee+, J. M. Jeon+, H. R. Kim, M. S. Kim, C. Ahn, S.-U. An, J. Choi, H. Kim, C. K. Jeong, H. Lin, **D.-H. Kang***, Y. M. Song*
22	A Van der Waals Optoelectronic Synapse With Tunable Positive and Negative Post-Synaptic Current for Highly Accurate Spiking Neural Networks	***Adv. Funct. Mater.*** (2026)	H. Yoon+, S. Park+, Y. K. Kim+, J. Baek, K. H. Kim, S. Yun, H. Son, J. Choi, B. C. Jang*, **D.-H. Kang***
21	*(Invited Review)* Spectrally Tunable 2D Material-Based Infrared Photodetectors for Intelligent Optoelectronics	***Adv. Funct. Mater.*** (2026)	J. Ha+, Y. Ma+, Y. N. An+, S.-U. An+, H. H. Jung, S.-T. Varjamo, J. Yoo, J. Min, H. Kim, F. Ahmed, S. H. Chae, Y. M. Song, W. Cai, T. Hasan, Z. Sun, **D.-H. Kang***, H.-J. Shin*, Y. Dai*, H. H. Yoon*

No.	Title	Journal	Authors
20	Avalanche-Enhanced Infrared Photodetection via Interlayer Absorption in a WSe2/MoS2 Heterostructure	*Nano Lett.* (2026)	B. Son+, H. Son+, Y. Kim, D. Kim, K. H. Kim, B. C. Jang, **D.-H. Kang***, D. Nam*
19	*(Invited Review)* 2D Materials in Logic Technology: Power Efficiency and Scalability in 2DM-MBC CFET	*Nano Lett.* (2025)	S. H. Shin+, **D.-H. Kang+**, H. H. Yoon+, J. Y. Park+, M. Song, H. Son, D. Ha*, H.-J. Shin*
18	*(Invited Review)* Enabling the Angstrom Era: 2D material-based multi-bridge-channel complementary field effect transistors	*npj 2D Mater. Appl.* (2025)	H. H. Yoon+, J. Y. Park+, Y. T. Megra+, J. H. Baek+, M. Song, D. Akinwande*, D. Ha*, **D.-H. Kang***, H.-J. Shin*
17	Polarity Engineering in a Single MoTe2 Device for Homogeneous Complementary Circuit Applications	*ACS Appl. Mater. Interfaces* (2025)	H. Son, S. Lee, S. Lee, H. Go, J. Baek, D. Kim, S.-U. An, J. Lee, H. H. Yoon, H.-J. Shin, J.-H. Lee, H. Lim, I.-M. Yi, M. Kim*, **D.-H. Kang***
16	Tactile neuromorphic system: convergence of triboelectric polymer sensor and ferroelectric polymer synapse	***ACS Nano*** (2023)	H. Kim, S. Oh, H. Choo, **D.-H. Kang***, J.-H. Park*
15	Pseudo-magnetic field-induced slow carrier dynamics in periodically strained graphene	***Nat. Commun.*** (2021)	**D.-H. Kang+**, H. Sun+, M. Luo+, K. Lu, M. Chen, Y. Kim, Y. Jung, X. Gao, S. J. Parluhutan, J. Ge, S. W. Koh, D. Giovanni, T. C. Sum, Q. J. Wang, H. Li, D. Nam*
14	A neuromorphic device implemented on a salmon-DNA electrolyte and its application to artificial neural networks	***Adv. Sci.*** (2019)	**D.-H. Kang+**, J.-H. Kim+, S. Oh, H.-Y. Park, S. R. Dugasani, B.-S. Kang, C. Choi, R. Choi, S. Lee, S. H. Park, K. Heo, J.-H. Park*
13	Rhenium Diselenide (ReSe2) Near-Infrared Photodetector: Performance Enhancement by Selective p-Doping Technique	***Adv. Sci.*** (2019)	J. Kim+, K. Heo+, **D.-H. Kang+**, C. Shin, S. Lee, H.-Y. Yu, J.-H. Park*
12	Rhenium diselenide (ReSe2) infrared photodetector enhanced by (3-aminopropyl) trimethoxysilane (APTMS) treatment	*Org. Electron.* (2018)	M. H. Ali+, **D.-H. Kang+**, J.-H. Park*
11	Highly sensitive and reusable membraneless field-effect transistor (FET)-type tungsten diselenide (WSe2) biosensors	***ACS Appl. Mater. Interfaces*** (2018)	H. W. Lee+, **D.-H. Kang+**, J. H. Cho, S. Lee, D.-H. Jun*, J.-H. Park*
10	*(Invited Review)* Recent progress in Van der Waals (vdW) heterojunction-based electronic and optoelectronic devices	*Carbon* (2018)	J. Shim+, **D.-H. Kang+**, Y. Kim, H. Kum, W. Kong, S.-H. Bae, I. Almansouri, K. Lee, J.-H. Park*, J. Kim*
9	Self-assembled layer (SAL)-based doping on black phosphorus (BP) transistor and photodetector	***ACS Photonics*** (2017)	**D.-H. Kang+**, M. H. Jeon+, S. K. Jang, W.-Y. Choi, K. N. Kim, J. Kim, S. Lee, G. Y. Yeom, J.-H. Park*
8	Poly-4-vinylphenol (PVP) and poly (melamine-co-formaldehyde)(PMF)-based atomic switching device and its application to logic gate circuits with low operating voltage	***ACS Appl. Mater. Interfaces*** (2017)	**D.-H. Kang+**, W.-Y. Choi+, H. Woo, S. Jang, H.-Y. Park, J. Shim, J.-W. Choi, S. Kim, S. Jeon, S. Lee, J.-H. Park*

No.	Title	Journal	Authors
7	Ultra-low doping on two-dimensional transition metal dichalcogenides using DNA nanostructure doped by a combination of lanthanide and metal ions	*Sci. Rep.* (2016)	**D.-H. Kang**, S. R. Dugasani, H.-Y. Park, J. Shim, B. Gnapareddy, J. Jeon, S. Lee, Y. Roh, S. H. Park, J.-H. Park*
6	A High-Performance WSe2/h-BN Photodetector using a Triphenylphosphine (PPh3)-Based n-Doping Technique	***Adv. Mater.*** (2016)	S.-H. Jo+, **D.-H. Kang**+, J. Shim, J. Jeon, M. H. Jeon, G. Yoo, J. Kim, J. Lee, G. Y. Yeom, S. Lee, H.-Y. Yu, C. Choi, J.-H. Park*
5	An ultrahigh-performance photodetector based on a perovskite--transition-metal-dichalcogenide hybrid structure	***Adv. Mater.*** (2016)	**D.-H. Kang**, S. R. Pae, J. Shim, G. Yoo, J. Jeon, J. W. Leem, J. S. Yu, S. Lee, B. Shin*, J.-H. Park*
4	Nondegenerate n-type doping phenomenon on molybdenum disulfide (MoS2) by zinc oxide (ZnO)	*Mater. Res. Bull.* (2016)	**D.-H. Kang**+, S.-T. Hong+, A. Oh, S.-H. Kim, H.-Y. Yu, J.-H. Park*
3	Controllable nondegenerate p-type doping of tungsten diselenide by octadecyltrichlorosilane	***ACS Nano*** (2015)	**D.-H. Kang**, J. Shim, S. K. Jang, J. Jeon, M. H. Jeon, G. Y. Yeom, W.-S. Jung, Y. H. Jang, S. Lee, J.-H. Park*
2	High-performance transition metal dichalcogenide photodetectors enhanced by self-assembled monolayer doping	***Adv. Funct. Mater.*** (2015)	**D.-H. Kang**+, M.-S. Kim+, J. Shim, J. Jeon, H.-Y. Park, W.-S. Jung, H.-Y. Yu, C.-H. Pang, S. Lee, J.-H. Park*
1	Indium (In)-and tin (Sn)-based metal induced crystallization (MIC) on amorphous germanium (α-Ge)	*Mater. Res. Bull.* (2014)	**D.-H. Kang**, J.-H. Park*

FULL-PUBLICATION LIST (CONTRIBUTING-AUTHORED)

Top 10% Journal (JCR)

No.	Title	Journal	Authors
27	Timing-Dependent Spiking Neural Network: Board-Level Hardware Implementation with Photoelectroactive Van der Waals Synapses	***Adv. Mater.*** (2026)	S. Kim+, J.-I. Cho+, S. Lee+, Y. Shin, J.-J. Lee, T. Jang, H. Kim, J. Oh, S. Lee, K. Ko, J. Kang, J. Lee, M. T. Flavin, **D.-H. Kang**, B. C. Jang, J.-H. Ahn, Y. Lee, S. M. Won*, J.-H. Park*, S. Oh*
26	Configurable anti-ambipolar photoresponses for optoelectronic multi-valued logic gates	*Appl. Phys. Lett.* (2024)	X. Cui, S. Kim, F. Ahmed, M. Du, A. C. Liapis, J. A. Mu\~noz, A. M. Shafi, M. G. Uddin, F. Ali, Y. Zhang, **D.-H. Kang**, H. Lipsanen, S. Kang, H. H. Yoon*, Z. Sun*
25	Triaxially strained suspended graphene for large-area pseudo-magnetic fields	*Optics letters* (2022)	M. Luo+, H. Sun+, Z. Qi, K. Lu, M. Chen, **D.-H. Kang**, Y. Kim, D. Burt, X. Yu, C. Wang, Y. D. Kim, H. Wang, Q. J. Wang, D. Nam*
24	Direct bandgap GeSn nanowires enabled with ultrahigh tension from harnessing intrinsic compressive strain	*Appl. Phys. Lett.* (2022)	D. Burt+, H.-J. Joo+, Y. Kim+, Y. Jung, M. Chen, M. Luo, **D.-H. Kang**, S. Assali, L. Zhang, B. Son, W. Fan, O. Moutanabbir, Z. Ikonic, C. S. Tan, Y.-C. Huang, D. Nam*
23	Controllable potential barrier for multiple negative-differential-transconductance and its application to multi-valued logic computing	***npj 2D Mater. Appl.*** (2021)	S. Seo, J. Koo, J.-W. Choi, K. Heo, M. Andreev, J.-J. Lee, J.-H. Lee, J.-I. Cho, H. Kim, G. Yoo, **D.-H. Kang**, J. Shim, J.-H. Park*

No.	Title	Journal	Authors
22	Biaxially strained germanium crossbeam with a high-quality optical cavity for on-chip laser applications	*Opt. Express* (2021)	Y. Jung[+], Y. Kim[+], D. Burt, H.-J. Joo, **D.-H. Kang**, M. Luo, M. Chen, L. Zhang, C. S. Tan, D. Nam*
21	Heterostrain-enabled dynamically tunable moir\'e superlattice in twisted bilayer graphene	*Sci. Rep.* (2021)	X. Gao[+], H. Sun[+], **D.-H. Kang**, C. Wang, Q. J. Wang, D. Nam*
20	1D photonic crystal direct bandgap GeSn-on-insulator laser	*Appl. Phys. Lett.* (2021)	H.-J. Joo[+], Y. Kim[+], D. Burt, Y. Jung, L. Zhang, M. Chen, S. J. Parluhutan, **D.-H. Kang**, C. Lee, S. Assali, Z. Ikonic, O. Moutanabbir, Y.-H. Cho, C. S. Tan, D. Nam*
19	Effect of large work function modulation of MoS 2 by controllable chlorine doping using a remote plasma	*J. Mater. Chem. C* (2020)	K. H. Kim, K. S. Kim, Y. J. Ji, I. Moon, K. Heo, **D.-H. Kang**, K. N. Kim, W. J. Yoo, J.-H. Park, G. Y. Yeom*
18	Polarity control in a single transition metal dichalcogenide (TMD) transistor for homogeneous complementary logic circuits	*Nanoscale* (2019)	J. Shim[+], S. woon Jang[+], J.-H. Lim, H. Kim, **D.-H. Kang**, K.-H. Kim, S. Seo, K. Heo, C. Shin, H.-Y. Yu, S. Lee, D.-H. Ko, J.-H. Park*
17	Ultrasensitive MoS2 photodetector by serial nano-bridge multi-heterojunction	***Nat. Commun.*** (2019)	K. S. Kim, Y. J. Ji, K. H. Kim, S. Choi, **D.-H. Kang**, K. Heo, S. Cho, S. Yim, S. Lee, J.-H. Park, Y. S. Jung, G. Y. Yeom*
16	Stable and reversible triphenylphosphine-based n-type doping technique for molybdenum disulfide (MoS2)	***ACS Appl. Mater. Interfaces*** (2018)	K. Heo[+], S.-H. Jo[+], J. Shim, **D.-H. Kang**, J.-H. Kim, J.-H. Park*
15	Layer-controlled thinning of black phosphorus by an Ar ion beam	*J. Mater. Chem. C* (2017)	J. W. Park[+], S. K. Jang[+], **D. H. Kang**, D. San Kim, M. H. Jeon, W. O. Lee, K. S. Kim, S. J. Lee, J.-H. Park, K. N. Kim, G. Y. Yeom*
14	*(Invited Review)* Electronic and optoelectronic devices based on two-dimensional materials: From fabrication to application	*Adv. Electron. Mater.* (2017)	J. Shim, H.-Y. Park, **D.-H. Kang**, J.-O. Kim, S.-H. Jo, Y. Park, J.-H. Park*
13	Efficient threshold voltage adjustment technique by dielectric capping effect on MoS 2 field-effect transistor	*IEEE Electron Device Lett.* (2017)	J. Park, **D.-H. Kang**, J.-K. Kim, J.-H. Park, H.-Y. Yu*
12	Extremely low contact resistance on graphene through n-Type doping and edge contact design	***Adv. Mater.*** (2016)	H.-Y. Park, W.-S. Jung, **D.-H. Kang**, J. Jeon, G. Yoo, Y. Park, J. Lee, Y. H. Jang, J. Lee, S. Park, H. Yu, B. S. Shin, S. Lee, J. Park*
11	Extremely large gate modulation in vertical graphene/WSe2 heterojunction barristor based on a novel transport mechanism	***Adv. Mater.*** (2016)	J. Shim, H. S. Kim, Y. S. Shim, **D.-H. Kang**, H.-Y. Park, J. Lee, J. Jeon, S. J. Jung, Y. J. Song, W.-S. Jung, J. Lee, S. Park, J. Kim, S. Lee, Y.-H. Kim, J.-H. Park*
10	Broad detection range rhenium diselenide photodetector enhanced by (3-aminopropyl) triethoxysilane and triphenylphosphine treatment	***Adv. Mater.*** (2016)	S.-H. Jo, H.-Y. Park, **D.-H. Kang**, J. Shim, J. Jeon, S. Choi, M. Kim, Y. Park, J. Lee, Y. J. Song, S. Lee, J.-H. Park*

No.	Title	Journal	Authors
9	High-performance 2D rhenium disulfide (ReS2) transistors and photodetectors by oxygen plasma treatment	***Adv. Mater.*** (2016)	J. Shim[+], A. Oh[+], **D.-H. Kang**, S. Oh, S. K. Jang, J. Jeon, M. H. Jeon, M. Kim, C. Choi, J. Lee, G. Y. Yeom, Y. J. Song, J.-H. Park*
8	Suppression of boron diffusion using carbon co-implantation in DRAM	*Mater. Res. Bull.* (2016)	S. H. Lee, S. G. Park, S. D. Kim, H.-C. Jung, I. G. Kim, **D.-H. Kang**, D. J. Kim, K. P. Lee, J. S. Choi, J.-W. Baek, M. Choi, Y. Park, C. Choi, J.-H. Park*
7	Improvement of the short channel effect in PMOSFETs using cold implantation	*Mater. Res. Bull.* (2016)	S. H. Lee, S. G. Park, S. H. Jeong, H.-C. Jung, I. G. Kim, **D.-H. Kang**, H.-J. Nam, D. J. Kim, K. P. Lee, J. S. Choi, W. Jung, Y. Park, C. Choi, J.-H. Park*
6	M-DNA/transition metal dichalcogenide hybrid structure-based bio-FET sensor with ultra-high sensitivity	*Sci. Rep.* (2016)	H.-Y. Park, S. R. Dugasani, **D.-H. Kang**, G. Yoo, J. Kim, B. Gnapareddy, J. Jeon, M. Kim, Y. J. Song, S. Lee, J. Heo, Y. J. Jeon, S. H. Park, J.-H. Park*
5	Phosphorene/rhenium disulfide heterojunction-based negative differential resistance device for multi-valued logic	***Nat. Commun.*** (2016)	J. Shim, S. Oh, **D.-H. Kang**, S.-H. Jo, M. H. Ali, W.-Y. Choi, K. Heo, J. Jeon, S. Lee, M. Kim, Y. J. Song, J.-H. Park*
4	Wide-range controllable n-doping of molybdenum disulfide (MoS2) through thermal and optical activation	***ACS Nano*** (2015)	H.-Y. Park, M.-H. Lim, J. Jeon, G. Yoo, **D.-H. Kang**, S. K. Jang, M. H. Jeon, Y. Lee, J. H. Cho, G. Y. Yeom, W.-S. J. Jung, J. Lee, S. Park, S. Lee, J.-H. Park*
3	Theoretical and Experimental Investigation of Graphene/High-k/p-Si Junctions	*IEEE Electron Device Lett.* (2015)	J. Shim, G. Yoo, **D.-H. Kang**, W.-S. Jung, Y.-C. Byun, H. Kim, W. T. Kang, W. J. Yu, H.-Y. Yu, Y. Park, J.-H. Park*
2	Germanium pin avalanche photodetector fabricated by point defect healing process	*Opt. Lett.* (2014)	J. Shim, **D.-H. Kang**, G. Yoo, S.-T. Hong, W.-S. Jung, B. J. Kuh, B. Lee, D. Shin, K. Ha, G. S. Kim, H.-Y. Yu, J. Baek, J.-H. Park*
1	n-and p-type doping phenomenon by artificial DNA and M-DNA on two-dimensional transition metal dichalcogenides	***ACS Nano*** (2014)	H.-Y. Park, S. R. Dugasani, **D.-H. Kang**, J. Jeon, S. K. Jang, S. Lee, Y. Roh, S. H. Park, J.-H. Park*

PATENTS

No.	Title	Country	Inventors
1	부성 미분 전달컨덕턴스 특성을 갖는 반도체 소자 및 그 제조 방법	*Korea* (2019)	박진홍, 심재우, **강동호**
2	Semiconductor Device with Negative Differential Transconductance and Method of Manufacturing the Same	*USA* (2020)	Jin-Hong Park, Jaewoo Shim, **Dong-Ho Kang**
3	인공 시냅스 소자 및 이의 제조방법	*Korea* (2021)	박진홍, 허근, 유광위, 구지완, **강동호**
4	Bipolar Optical Synaptic Device	*USA* (2026)	**Dong-Ho Kang**, Hyejin Yoon, Soeun Park
5	양극성 광 시냅스 디바이스	*Korea* (2026)	**강동호**, 윤혜진, 박소은

INVITED TALKS

No.	Title	Conference	Authors
8	*(Invited)* A Self-Inhibitory WSe2 Retinomorphic Transistor for Adaptive In-Sensor Processing	*The 13th Korean Symposium on Graphene and 2D Materials* Busan, South Korea 2026.07.09 - 07.11	**D.-H. Kang***
7	*(Invited)* Optoelectronic Synaptic Devices for in-Sensor Computing Applications	*The 22nd International Symposium on the Physics of Semiconductors and Applications (ISPSA 2026)* Jeju, South Korea 2026.06.29 - 07.02	**D.-H. Kang***
6	*(Invited)* Van der Waals Heterostructure-Based Optoelectronic Devices for Neuromorphic and In-Sensor Applications	*한국전기전자재료학회 2026 하계학술대회* Busan, South Korea 2026.06.25 - 06.27	**D.-H. Kang***
5	*(Invited)* Ferroelectric phototransistors for in-sensor processing toward all-day facial recognition	*10th IEEE Electron Devices Technology and Manufacturing (EDTM)* Penang, Malaysia 2026.03.01 - 03.04	**D.-H. Kang***
4	*(Invited)* 인센서 컴퓨팅 응용을 위한 광-전자 시냅스 소자 연구	*2025 Photonic Conference* Pyeongchang, South Korea 2025.11.26 - 11.28	**D.-H. Kang***
3	*(Invited)* TMD Photodetector: Performance Enhanced by Chemical Doping Technique	*14th Asia-Pacific Conference on Near-field Optics* Busan, South Korea 2023.06.19 - 06.22.	**D.-H. Kang***
2	*(Invited)* Pseudo-magnetic fields in strained graphene for building optical quantum computing systems	*7th IEEE Electron Devices Technology and Manufacturing (EDTM)* Seoul, South Korea 2023.03.07 - 03.10.	**D.-H. Kang***
1	*(Invited)* A Study of Carrier Dynamics in Pseudo-Landau-Quantized Graphene	*한국광학회 2023 동계학술대회* Busan, South Korea 2023.02.14 - 02.17.	**D.-H. Kang***